SECURE CLOUD STORAGE APPLICATION FOR CLOUD PLATFORMS

B.TECH FINAL YEAR PROJECT REPORT

SUBMITTED BY

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UNDER THE GUIDANCE OF

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In partial fulfillment for the award of the degree
Of
Bachelor of Technology
in
Information Technology

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MAY, 2022

With the blessings of Almighty. To
My beloved parents, family members, and
My mother

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SUPERVISOR'S CERTIFICATE

This is to certify that the work in the thesis/report entitled, ''**SECURE CLOUD STORAGE APPLICATION FOR CLOUD PLATFORMS**'', submitted by **Shikhar Mishra** Bearing **Roll No B418048** to IIIT Bhubaneswar, is a record of an original research work carried out by him under my supervision and guidance during 8th Semester in the partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Information Technology Engineering of the University. Neither this report nor any part of it has been submitted for any degree or academic award elsewhere.

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We the below signed, after checking the thesis/report mentioned above and the official record book (s) of the student, hereby state our approval of the thesis/report submitted in partial fulfillment of the requirements of the degree of Bachelor of Technology in Information Technology at International Institute of Information Technology, Bhubaneswar. We are satisfied with the volume, quality, correctness, and originality of the work.

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DECLARATION OF ORIGINALITY

I, SHIKHAR MISHRA, bearing roll number B418048, hereby declare that this thesis/report entitled "**SECURE CLOUD STORAGE APPLICATION FOR CLOUD PLATFORMS**", presents my original work carried out as a bachelor student of International Institute of Information Technology, Bhubaneswar, and to the best of my knowledge, contains no material previously published or written by another person, nor any material presented by me for the award of any degree or diploma of International Institute of Information Technology, Bhubaneswar or any other institution. Any contribution made to this work by others, with whom I have worked at International Institute of Information Technology, Bhubaneswar or elsewhere, is explicitly acknowledged in the thesis/report. Works of other authors cited in this dissertation have been duly acknowledged under the sections "Reference". I have also submitted my original research records to the scrutiny committee for evaluation of my thesis/report.

I am fully aware that in case of any non-compliance detected in future, the Senate of International Institute of Information Technology, Bhubaneswar may withdraw the degree awarded to me on the basis of the present dissertation.

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List of Acronyms

AES	(Advanced Encryption Standard)
SCSA	(Secure Cloud Storage Application)
HTML	(Hyper Text Markup Language)
CSS	(Cascading Style Sheets)
RSA	(Rivest-Shamir-Adleman)

Abstract

Cloud storage has proven to be a boon to many businesses and people as it spares users of the responsibility of information protection and safety while also cutting infrastructure costs. It has also proved its enormous strength and universal acceptability, which is important for cloud computing's rapid expansion. The cloud service provider ensures that the data privacy and confidentiality remain protected in addition to providing storage capacity. For saving, modifying, and exchanging individual and organizational information, Google Drive as well as other cloud-based document storage systems are frequently utilized. Managing data security and access control is among the most challenging aspects of using online storage. As a consequence of management inefficiency and malevolent attacks, major security incidents continue to happen, resulting in enormous volumes of confidential data leakage at the cloud storage tier. As a result, in order for large-scale cloud computing to work, a comprehensive online storage solution is required.

As a result, we present a novel method for developing a safe cloud storage software for Dropbox, Google Drive, Office365, and other similar services. We also create an access control system for this platform that enables users to be added or removed. The main goal of the project is to construct a program that links to a cloud based storage platform & allows users to upload information whilst also encrypting it, then decode that encrypted file while they receive it. The program can manage multiple users signing in and downloading password-protected data, and also the key is transmitted. We also see its execution and examine its functioning effectiveness and look out for further activities that can be done in future.

Keywords: Secure Cloud Storage, AES, Google API, Fernet.

1. Introduction

As information technologies and the Internet both progressed, cloud technology has risen in relevance, and so it is becoming one of the biggest key aspects of research in recent decades. The computational operations are split up throughout a huge number of computational resources inside a pool of resources. Cloud computing is just the usage of computational web services. From basic infrastructure until complete SaaS (Software as a Service) capabilities, these services comprise it all. Cloud computing, that has a large amount of processing resources, allows users to easily access resources provided by the cloud. Thus, it makes cloud computing extremely scalable, accessible, and sustainable, while also giving consumers a great deal of flexibility. It Allows users to store essential information in the cloud and access it via the Web from just about anywhere in the globe, eliminating the need to worry about device or disc malfunctions, for instance. This reduces any need for businesses to invest in in-house facilities, freeing up resources to focus on enhancing and expanding their business. The cloud service also allows for on-demand compute and automatic scaling. As an outcome, consumers can scale their processing needs back and forwards as needed. The majority of individuals nowadays have exposure to cloud-based channels such as email, gaming, videos, images, and files. There's also a large number of services to pick from in the cloud. For such a wide variety of computer demands, cloud - based solutions are increasingly becoming the favorite option of businesses. Because of its versatility, cloud computing has become an appealing option for both organisations and individuals.

Cloud Data storage, a traditional Infrastructure as a Service (IaaS) commodity, is one of the most basic and important cloud web services. As a consequence to advancements in technologies and the internet, information is booming. As an

outcome, corporations and research scholars are becoming much more interested in cloud storage technology created in combination with cloud computing. With online storage, information is stored on numerous third-party servers rather to the data centers utilised in traditional networked storage of information. While storing the data, the user can view a virtual server, that creates the impression as if the information is maintained in a certain address with a specific title. However, in actuality, the location does not exist. It's merely a make-believe term for a cloud-based virtual territory. In reality, the customer's information might be kept on either of the cloud's systems. Regardless of the fact that the address is fictitious, the user perceives a physical place for his information and has access to it as though it had been attached with his own machine. A client will send copies of documents through the web to a data server, which records the information. When a client requires data, he or she connects with the data server via a web-based interface.

Cloud storage (Storage as a Service) has now become the quickest developing field of cloud computing due to its high scalability, transparency to clients, and versatility of distribution on demand. Storage as a service comes to the fore and acts as the foundation of many systems, including pattern classification, image forensics, and fraud detection. As an outcome, the cloud would indeed be capable of handling greater data volumes. In the cloud sector, programs such as Google Drive, Swift, Dropbox, and Amazon Web Service (AWS) have established the de facto standard.

However, there are also several challenges with cloud storage that really should be resolved before it can live up to its potential. Two among the most serious issues about cloud storage are data security and privacy. Putting aside the advantages of cloud computing, businesses face significant security challenges when shifting applications as well as confidential material to and from the cloud, including privacy, data theft, and identity theft protection. In 2012, for example,

a massive data loss incident involving Google Gmail mailboxes occurred, leading to the loss of data for around 1.6 million Google mailbox customers. In May 2013, hackers gained access to Amazon's cloud computing services, revealing more than 259 million LinkedIn users' personal information. Within certain circumstances, Google Drive did have a flaw that just might allow unapproved persons exposure to a subset of data, according to Google. While delivering useful features, the Openstack Swift methodology has a number of serious security flaws. The OpenStack Swift technology frequently retains data in a plaintext form for performance reasons. As a consequence, unauthorised access of user information just at storage layer can occur. As a result, it's critical to keep confidential material against leaking outside of corporate on-premises systems and onto insecure clouds.

Every one of these responses demonstrates that one of the most significant barriers to realising the full potential of online storage involves privacy and security, which now has caught considerable interest of academics and industry throughout the world. Individual files and health information, as well as financial information from enterprises, are especially susceptible data stored in the cloud. If these challenges cannot be addressed, the advantages of online storage will indeed be diminished. Most firms, for instance, would be unable to maintain sensitive information over an unprotected infrastructure or in a place in which the CSP is down.

Researchers have suggested a number of strategies and procedures to maintain classified information safe and secure, but nobody has proven to be particularly effective. The most common advice would be to encrypt information prior to uploading it all to the cloud, and physically retrieve and decrypt it as necessary. This operation, if performed manually, would not only be exhausting, but it would also consume a considerable amount of time and resources, reducing productivity levels in larger organisations. In 2007, Bethencourt became the first to use

Ciphertext Policy Attribute Based Encryption(CP-ABE). In the CP-ABE mode, the ciphertext as well as the "access policy" are linked, and the customer's "private key" is associated to the "attribute set." This approach, on the other hand, requires an existence of at least one absolutely trustworthy characteristic or centralized administration. The central authority's betrayal in this case will jeopardise the overall structure. As a response, establishing a mechanism/application that effectively integrates data security and privacy is vital to assuring the security of sensitive data and safeguarding the privacy of trusted users. This is exactly what this study recommends. I introduce a secure and effcient cloud storage program that offers a trustworthy computing platform for cloud computing environments by utilising a mechanism that encrypts data on the client side prior to sending them to the cloud and decodes them after retrieving them. The data is encrypted and stored in the cloud, making it inaccessible to someone who does not have the key. The following section talks through it in further depth.

2. Motivation & Objetcive

2.1 Motivation

As said in the preceding sections, developing a secure cloud storage architecture whereby consumer can safely keep information and data of every sort and exchange it with other users whom they desire to share it with using a secret key is critical. This type of system will support clients in managing their data in the cloud while also allowing them to block unauthorised access to their data, from even the CSP. This technology will assist businesses in operating more successfully and will enable them to fully utilise cloud storage. This fueled my desire to complete this endeavour.

2.2 Objective of the Project/Research

The goal of this project is to provide a secure cloud storage solution for services like Dropbox, Amazon Web Services, Google Drive, Office365, and many others. The proposed model intends to generate a trustable cloud infrastructure for cloud computing systems by providing a mechanism that encrypts files at the client end using a secret key before transferring it to cloud memory and decrypts it using the same secret key after retrieving it all from cloud storage, as well as restricting data access to unauthorized users. Both of these steps are carried out on the client end and need the use of a private key. As a consequence, the secret key never leaves the machine, and the client can be assured that data stored in the cloud is secure. The application will also be able to handle multiple people signing in and accessing these encoded files at the same time, with the key being shared among them. The most crucial component of the strategy is determining how to reliably distribute symmetric keys to genuine group members. For instance, all documents transferred to the cloud will be encrypted, and only users of the "Secure Cloud

Storage Group" would be able to decrypt them. For all other users' info will be encrypted. We create and install a key management system for the app that allows you to securely share files and add or remove users from your group.

- This project's authorised cloud storage is Google Drive. The framework is structured with the Python programming language, as well as the user experience interface is built with HTML, CSS, and JavaScript. The cryptography technique we use is known as Fernet. It uses AES in CBC mode with a 128-bit key for encoding data and HMAC with SHA256 for identity verification. The backend is written in Python. The following are some of the advantages of this architecture:
- the ability to add and remove numerous users at the same time.
- The information is encrypted while uploading and decoded after downloading.
- The interface design is straightforward.
- Users won't be able to intercept the files unless you know the key.

2.3 Organization of the Report

- Chapter 1 includes Introduction to the project.
- Chapter 2 includes Motivation & Objetcive of the project
- Chapter 3 includes Research Plan and Timeline
- Chapter 4 includes Related Work and Literature Survey.
- Chapter 5 includes Proposed System and Implementation.
- Chapter 6 includes Result Analysis.
- Chapter 7 includes Conclusion and Future Work.
- Chapter 8 includes References.

3. Research Plan and Timeline

Sl. No.	Activity	First 45 Days			Second 45 Days			Third 45 Days		
		15	30	45	60	75	90	105	120	135
1.	<Data Collection and Literature Survey>									
2.	<Data Analysis and processing>									
3.	<Modelling the System Architecture>									
4.	<Developing Algorithm for the Model>									
5.	<Implementation>									
6.	<Testing and validation>									
7.	<Further Improvements and Final System Delivery>									

4. RELATED WORK AND LITERATURE SURVEY

In Cloud computing, there are indeed a variety of important policies to consider, including anonymity, confidentiality, protection, liability, and reliability. Cloud computing security always has been a critical component of cloud service providers' overall service quality. The safe exporting of confidential and business-critical information and operations is the key challenge implicit inside the cloud computing paradigm. The secrecy and authenticity of cloud - based data activities is another key concern in cloud computing. According to the Cloud Security Alliance's top hazard case study report, due to management ineptitude and aggressive attacks, two-thirds of incidents would result in client loss of data. Data manipulation as well as other risks associated with the process manipulation can also be mitigated by implementing a secure storage solution that encodes any sort of information utilizing cryptographic methods, whether it's a document or an image.

One of most popular types of encryption protocols used for encoding and decoding include symmetric key technique, asymmetric key technique, and combination key methodology. Information and key encryption will safeguard data privacy and establish a secure cloud network. The process of turning a document or data into an unintelligible form which an invader cannot comprehend is known as encryption. The client has plain text, which is encrypted into an unreadable format using one of the cryptographic protocols, known as cypher text. When the target receiver receives the message, it can also be decoded into original message. To protect data security, encryption is most widely employed in network communications.

4.1 Symmetric Encyption

Since we only need one secret key for encryption and decryption of data, this is among the most basic encryption approaches. Block cyphers like DES, Blowfish, AES, Camellia, Serpent, plus stream cyphers like FISH, Py,RC4, QUAD, SNOW, and others are among the symmetric key encryption methods in use.

4.2 Asymmetric Encryption

In asymmetric encryption algorithms known as public key cryptography, two separate keys are used for encryption and decryption. The public key is used for encoding and is accessible to the public, whereas the secret key is used for decryption.

The most often used cryptography algorithms for cloud computing are as follows:

- **AES :** In the field of cryptography, the Advanced Encryption Standard (AES) is perhaps the most extensively used symmetric-key encryption technique. AES is a block cypher having a 128-bit block length as well as 128, 192, and 256-bit key sizes, correspondingly. AES has three different key lengths: 128, 192, & 256 bits. 128-bit keys needs ten rounds of computation for encoding and decoding, 192-bit keys take 12 rounds, and 256-bit keys demand 14 rounds. As for the last round in every case, the encryption / decryption phases are identical. Both hardware and software become more efficient as a result.
- **MD5:** MD5 (Message-Digest algorithm 5) is indeed a frequently used and also well-known hash function which generates a 128-bit hash result. MD5 is a widely utilised protection mechanism in a wide range of applications. It is often used to ensure that files aren't corrupted. The MD5 value generated for a document is thought to be a reliable fingerprint which can be used to confirm that document's authenticity. This algorithm was

created in C, Perl, and Java, among other computer languages. The transmitter encrypts the data with the recipient's public key, and the receiver decrypts it with its private key in the MD5 algorithm.

- **DES:** The DES algorithm is among the most extensively used encryption techniques (Data Encryption Standard). DES is just a symmetric block cypher invented by IBM. DES uses a 56-bit key to encode and decode a 64-bit data block. The technique is best suitable to be performed on hardware instead of software due to the low throughput and time consuming nature.
- **RSA:** A public key and a private key are used in the RSA algorithm, which is also referred as public-key cryptography. The public key, that can be utilised to encrypt the message, is accessible to anyone. Communications can be decrypted using the private key. Before even being stored, information can be encrypted, and secure communication pathways can be constructed.

The concepts in this paper were partly inspired by and partially based on the recent work by renowned academics in the realm of secure information storage techniques. Storer et al. contributed the use of convergent encoding and a verification framework for authorised and anonymous access. The study does not offer a solution to the vulnerabilities with convergent encryption.

Shah et al. proposed a cloud-based data protection storage method for the Apache Spark architecture, which prevents data leaks and increases Apache Spark security. Zerfos et al. created a Hadoop-based safe distributed storage method that gives cloud data security via information dispersion and encoding. It performs data decoding and assembly before reading data. To safeguard keys from being hijacked, this solution necessitates the use of a key cache server, and then all keys should then be kept in storage only. To handle the key, several approaches engaged the services of a third-party. The trustworthiness of third parties is

assumed. The assumptions cloud doesn't really necessarily exist in the real world cloud storage setups.

Orner K. Jasim Mohammad, SafiaAbbas, EI-Sayed M. EI-Horbaty suggested a Comparative Study on Latest Cryptographic Algorithms Based On The cloud Computing System. The variations among current encryption techniques utilised in cloud computing are compared in this study (DES, 3-DES, AES, RC4). According to studies, the AES encryption technique has some benefits over all the others, especially in terms of encryption speed. In addition, the AES algorithm has also been proven to be resistant to hacking attempts. D. S. Abdul. Elminaam published A Performance Evaluation of Symmetric Key encryption Technologies AES, DES, 3DES, RC2 and RC6. Varying parameters for each encryption scheme were tested, including multiple data block sizes, diverse data types, battery energy consumption, key sizes, and cryptographic performance. Whenever the packet size is increased, there is no discernible difference between the results when shown in hexadecimal or base 64 encoding. It was discovered that AES operates better when key sizes, batteries, and time consumption are changed. In light of this, we chose AES for the suggested system.

5. Proposed System and Implementation

5.1 Requirements Analysis

Data encryption is commonly used in communication-related activities to prevent the information of text, messages, and sometimes even files from it being read by anybody other than the intended targets. Aside from that, it enables recipients to verify that a response came from a specific intended recipient and was not apprehended and tampered with. Public key cryptography is frequently used in end-to-end encryption. Under typical situations, Symmetric Encryption can be used to accomplish this.

The suggested mechanism's major goal is to protect cloud storage from data breaches caused by targeted attacks (e.g., disc cloning) or administrative carelessness (e.g., misconfiguration), in the event that attackers or a malevolent administrator are able to take away user data. In many circumstances, information dispersion or encoding is the most widely used method to achieve this purpose. Both strategies have the potential to protect privacy, but they will also have risks. By encrypting data with cryptographic keys, data encryption technique store information in cypher texts. However, by stealing the keys, hackers can potentially recover the actual data. This poses the issue of key management and preservation. As a result, in order to maximise the anonymity of cloud storage systems, the suggested scheme should fully use the method's benefits while successfully managing its drawbacks. Meanwhile, the mechanism's increased cost should remain within a fair range.

Specifically, following properties should be met:

- Property 1: Any user data kept inside the cloud will not be disclosed, seen, stolen, or utilised by unauthorized parties, such as hackers or malicious administrators, from the standpoint of maintaining cloud data confidentiality.
- Property 2: Any parameters linked to keeping cloud data secure, such as cryptographic keys, should be protected based on property 1.
- Property 3: The extra performance overhead of installing the proposed technique should be acceptable to the user.

It was discovered that AES operates better when key sizes, batteries, and time consumption are changed. In light of this, we chose AES for the suggested system.

5.2 Proposed System Overview

We present SCSA, a Secure Cloud Storage Application, to accomplish the primary aim and properties mentioned, as well as to address issues such as data breach and data manipulation. Our main goal is to build a credible computing environment for cloud computing systems by providing a framework that encrypts files at the client side with a secret key before transferring it to cloud memory and decrypts it with the same secret key after retrieving it from online storage, along with restricting access to data from non-authorized users. The system also offers a multi-user capability that allows the admin to control numerous users at the same time. Using the suggested system, users can upload information to the cloud in an encoded format, ensuring that the information contained therein stays safe and secure, and then download it and receive it in its original version when needed. After downloading, the system decrypts the document/file and sends it to the user. A secret key is used for both encoding and decoding (symmetric-key algorithm). This aids users and organisations in keeping their files safe in the cloud. Another significant feature it offers is the

ability for the system to have numerous users, and the file may be distributed with the required users by exchanging the secret key with them. The Administrator does have the ability to construct usernames and passwords, as well as distribute the document with the needed user(s) by sharing the secret key with them. Dropbox, Amazon Web Services (AWS), Google Drive, Office365, and other Cloud Services (CSP) can all be integrated with the secure cloud solution.

One crucial assumption made during the creation of the programme is that the cloud storage group's activities are always monitored and its contents are always available to others. As an outcome, the information of the online storage are encoded for data security, and only users with knowledge of the symmetric keys can decrypt them. Both procedures of encryption and decryption are carried out by the client. As a result, the secret key never leaves the machine, and the user can be assured that data stored in the cloud is secure. With the key shared between users, the system will be able to support several people signing in and obtaining these encrypted files. The most crucial component of the design is determining how to reliably distribute symmetric keys to genuine group members. For instance, all documents transferred to the cloud will be encrypted, and then only users of the "Secure Cloud Storage Group" will be able to decrypt them. For, all other users' info will be encrypted. In addition, the system contains a key management system that enables users to safely transfer documents and add or delete users from their groups. The Administrator is responsible for the generation of usernames and passwords, as well as for adding and deleting users from the group. Fernet is the encryption technology utilised in this, which is based on the AES encryption algorithm in CBC mode for encrypting and decrypting, as well as HMAC using SHA256 for authentication.

5.3 System Implementation

There are three main elements in the system:

The user, the storage system layer, and the cloud service provider (CSP).

- User - A user is a customer, a consumer, or an organisation that wishes to maintain data or information in the cloud. For data storage and processing, the User submits to cloud service suppliers' storage and computational resources. The proposed approach allows the user to access its Cloud Storage service by his own machine.
- Storage System layer – The Storage System Layer is the proposed method which connects with the Cloud Service Provider and allows users to upload/download encrypted/decrypted files from the cloud. User creation, file uploading and downloading, document sharing, and data security are all aspects of this layer, which is deployed over the client's system.
    1. User Generation: Each user has his or her own account. As a result, each user must first get a unique account before using the cloud resources. The administrator just needs to perform this once to set up an account with a username and password. Then the user (she or he) can use the username and password to connect into the system from anywhere and upload/download documents utilising the file upload and download function.
    2. File Uploading and Downloading: The user can upload files from almost anywhere using his or her account and password. Later, she or he can use the same key to retrieve the data. Before being recorded in the repository, the information of the file will be encoded using AES encryption. When the content is downloaded, it will also be decrypted. As a result, the chances of an assault during upload/download period are lowered.

3. File Sharing: When necessary, the application also allows users to share files with other members of the group. This is accomplished by providing them with the document's secret key. Multiple people can access the data at the same time using the application.
4. Data Security: The data uploaded by the user is encrypted with 128-bit AES encryption. AES is a symmetric encryption technique that is quick and efficient.

- Cloud Service Provider (CSP) - The CSP is an organization that has a set quantity of storage capacity and computer power. Users can access CSP's storage or computational services. Google Drive is the project's approved cloud storage.

The schema of the proposed system includes the following classes/algorithms which are used to implement the all features:

Main file,Users, GoogleDriveAPI_Interface, MenuTypes, Encryption and Auth

1) Main file – It is the setup's primary class, which is invoked at the start of execution. This is what will be used to generate the user's front end. It's small and simple, and it uses the Users algorithm for login and the MenuType method for menu generation.

When the application initially starts, it checks to see whether a key exists; if not, one will be automatically generated. The programme then checks to see if any previous users have registered. If there are no users, the user is prompted to make an administrator account, which has the ability to create new users and produce new keys. Otherwise, the user would be prompted to log in. If you sign in as an administrator, you'll get a menu with additional features tailored to that user. If you don't, you'll be provided with a regular menu. This runs indefinitely to allow people to sign in and out whenever they want.

2) Users Algorithm –

This is where usernames & passwords are generated, erased, and saved, as the name implies. This is a very essential algorithm that is used for a variety of purposes. This algorithm is made up of several subset functions –

Save_list (obj, name) – This method is used to pickle (store) the information of users that are currently participating in the programme. It stores the user's id and password like a key value pair in a dictionary. It is invoked whenever a new user is generated.

load_list(name) - This method loads the object that contains the users and passwords. Whenever the user class is called, it checks to see if the usernames are correct.

newUser() - To create a new user, this function is used. It's straightforward: it asks for a user name, alerts the user if the user name already exists, and then requests for a password, which it puts in the dictionary & pickles.

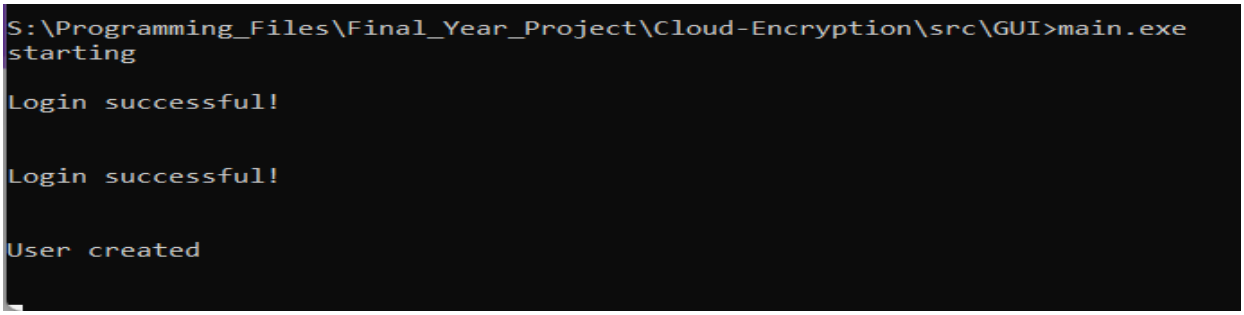

Figure 1: User Generation and the backend terminal

Login() - This method is invoked in the main file and displays a login page to the user. It verifies if the user logging in is an admin, a standard user, or if they fail to login three times, the session will be terminated.

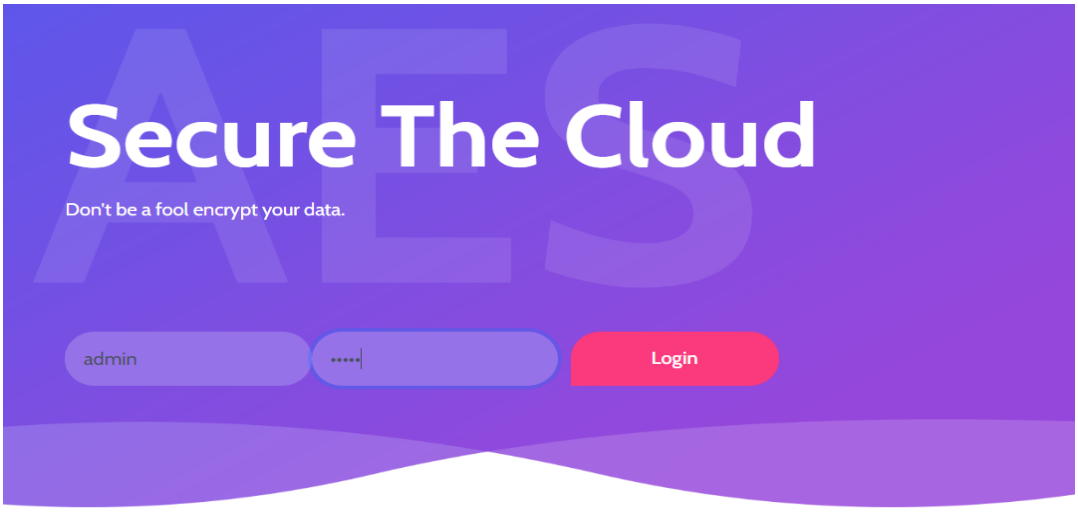

Figure 2: Login Screen

deleteUser() - The administrator uses this to delete a user. It prevents the administrator account from ever being deleted and notifies the user if somehow the user does not exist. When a user is deleted, the username dictionary is pickled and safely saved once more.

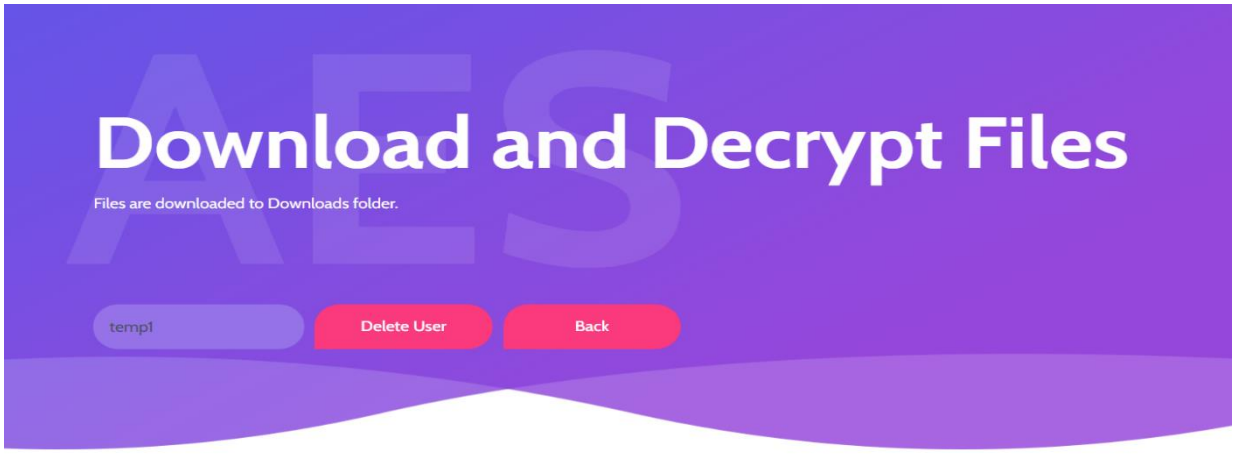

Figure 3: User Deletion

3) GoogleDriveAPI_Interface –

It serves as a link between the software and Google Drive, allowing the scheme to do the necessary download and upload tasks. It has all of the necessary functionality for uploading, downloading, and searching for files. Among the functions are:

• **uploadFile(fileName)** – This function is used to upload file over the cloud storage (Google Drive).

• **downloadFile(file_id, fileName)** - This function is used to download file from the cloud storage (Google Drive).

• **searchFile(query)**

4) MenuTypes Algorithm –

This is the method that generates menus for privileged (administrator) and non-privileged (ordinary) users. To perform essential functions, it uses the majority of classes and algorithms. Among the functions are:

PrivilegeMenu() - This creates a menu for the administrator that includes extra options like adding and deleting users as well as producing a new key. The majority of activities are conducted here, such as uploading a file, which will be encrypted before being uploaded, and downloading a file from the disc, which will be downloaded and decrypted.

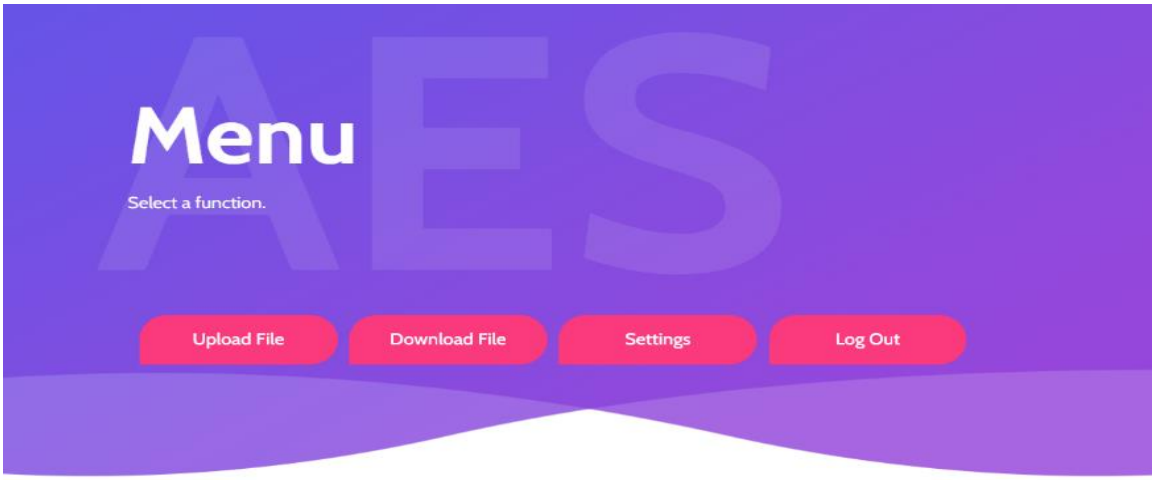

Figure 4: Administrator Menu

StandardMenu() - This is the panel that is displayed to the average user. This menu is largely identical to the Administrator Menu above, with the exception that the settings choice has been deleted to prevent the user from doing privileged operations.

5) Encryption Algorithm –

This algorithm, as the name implies, performs encryption and decryption as well as key generation. Fernet is the name of the encryption technology employed in this system. For encryption, it employs AES in CBC mode with a 128-bit key, and for authentication, it uses HMAC with SHA256.

Why AES encryption?

A referenced research paper compares cryptography algorithms DES, AES, and RSA based on the key length, cypher type, block size, safety, ease of hardware and software implementation, encryption/decryption performance, and other factors. Based on the experiments, the study discovered that the AES algorithm uses the least amount of encryption time while the RSA algorithm spends the most. Based on their findings, they concluded that the AES encryption is far superior to the DES and RSA algorithms. It is clear that RSA is the least safe, whereas AES is the most reliable and fastest. The fastest and most trustworthy distribution of services to its clients is a major issue for all organisations and providers. As a result, the AES algorithm is employed to encrypt and decrypt data since it is quicker and more safe than other algorithms.

Features	DES	AES	RSA
Developed	1977	2000	1977
Key Length	56 bits	128,192,256 bits	More than 1024 bits
Cipher Type	Symmetric block cipher	Symmetric block cipher	Asymmetric block cipher
Block size	64 bits	128 bits	Minimum 512 bits
Security	Not secure enough	Excellent secured	Least secure
Hardware & Software Implementation	Better in hardware than software	Better in both	Not efficient
Encryption and Decryption	Moderate	Faster	Slower

All the functionalities of this algorithm are divided into subset functions:

KeyGen() - KeyGen is a programme that generates a new key and stores it on the computer.

KeyRead() - This is used to extract the key out from computer; if there isn't one, it will produce one automatically by invoking KeyGen ().

Encrypt(filename, Key) - This is needed to encode the document, which is accomplished via Fernet (which utilises AES encryption).

Decrypt(filename, Key) - This is used to decrypt the file that had been encrypted.

6) Auth Algorithm – When calling APIs, this method is used to validate the Google Drive user. This is derived from the QuickStart tutorial for the Google Drive API.

GUI

A smooth graphical user interface (GUI) is combined with it to enable convenience of functioning and to make the service easy to use. This was accomplished with Python's Eel module, which converts Python functions to JavaScript and vice versa. This aids in the development of a web app with an HTML, CSS, and JavaScript user interface but a Python backend. The console (terminal) also runs in the background for demonstration purposes, allowing you to see what's going on in the back end.

Figure 5: GUI

6. Result Analysis

To test how well the proposed system (SCSA) works, I deployed it on my own system and assessed it at each step to see if it met all of the requirements and functioned effectively. The system specifications are - Intel(R) Core (TM) i5-8250U CPU with 8 GB RAM and Windows 11 Home Operating System. The following areas of SCSA' performance are studied and evaluated: User Creation and Deletion, File Encryption and Decryption, Uploading and Downloading files from Google Drive, Supporting Multiple Users, and Time Complexity. For the analysis we use an Admin account with user id – admin and password – admin.

Starting with **User Creation and Deletion**, this task can be performed by admin. After entering inside the system, using the administrator account, the option to add and delete users is used. To add a user, the admin goes to Add User, provides the required id and password and clicks on the Add User button, and the user gets created. The message "User Added" appears on the terminal upon successful addition, Similarly, to delete a user, select Delete User and then type the user's name. If the user exists, it will be erased, and the terminal will display "User Deleted.". For testing we created a user temp1 and then tried deleting it. It was seen that both these operations were executed successfully. SCSA also allows organisations to have multiple users and is capable of handling multiple users at the same time. User creating is done by the administrator and he can create as many users as required.

Moving on to the next step, which is to **see if the file was encrypted when it was uploaded and if it was decrypted when it was downloaded**. For this, we try to upload and then download a file for this. A pdf document was chosen, and the Upload Data feature was used to upload it. Following the upload, the

google drive was examined to determine if it had been successfully uploaded. The file had been uploaded in an encrypted manner, and when manually opened, the data was totally encrypted, making it incomprehensible. We then used the Download Data function to download the file, which successfully downloaded the pdf file and decrypted it, restoring it to its original state.

Time Complexities Analysis — Because SCSA uses 128-bit AES encryption, the time it takes to encrypt a file is proportional to its size. It encrypts an N-byte file in O(N). As a result, the time it takes to encode a file is exactly related to its size, and on average, the time cost is very little.

We can conclude that the proposed mechanism, SCSA, accomplishes exactly what a user would expect it to do and has the following merits after assessing all parts of it:

- Users can be added and removed in bulk.
- Information is encrypted as they are uploaded and decrypted after they have been downloaded.
- The user interface is simple to use.
- You can't decrypt files unless you have the key.

Along with the merits, there are a few aspects which can improved in the future. This includes the requirement to physically share the key in order to avoid someone else from gaining it. Furthermore, the GUI is created in a single file and it is not properly secured. Anyone with a basic understanding of CSS and the Chrome website development tools may enable admin-only features.

7. Conclusion & Future Work

7.1 Conclusion

Consumers and businesses can use software without installing them and manage their important documents via the internet thanks to cloud computing. There are a number of key security concerns with cloud computing technologies, including confidentiality, secrecy, telecommunications capacity, and dependability. However, the most pressing issue among these concerns is ensuring and providing a secure workplace. This study proposes a Secure Cloud Storage Application (SCSA) to address this problem. The goal of this solution was to provide a secure cloud storage solution for services such as Dropbox, Google Drive, and others by implementing a mechanism that encrypts data at the client end with a secret key before transferring them to the cloud and decrypts them with the same secret key after retrieving them from cloud storage. The application can also support numerous users at the same time. The application's framework is Python, while the user interface is HTML, CSS, and JavaScript. The cloud service used in this is Google Drive and was connected using Google Drive API. AES encryption was utilised in this, which has shown to be the most secure of all encryption schemes. SCSA can effectively prevent user data at the cloud storage layer, according to the results, and the time complexity is also acceptable.

7.2 Future Work

The research's long-term goal is to expand this cloud storage paradigm and address certain limitations. This secure cloud storage architecture can be modified and extended to include a method of sharing the key by decrypting the private key with a public key, resulting in a hybrid key of sorts. To improve the security of encrypted documents, a process can be created that updates both the ciphertext and its key at regular intervals. A verification method can also be introduced that checks the integrity of the data saved on the cloud server on a regular basis, which will help keep the CSP under control. More security in the GUI is also required to ensure that only administrators have access to the privileged menu.

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